

Phase 1: Core Electrostatics

The Physics of Stationary Charges

Master Study Guide & Step-by-Step Derivations

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1 Coulomb's Law and Linear Superposition

1.1 The Fundamental Axiom

The entire field of electrostatics is built upon one single experimental observation: **Coulomb's Law**. Coulomb observed that if you take two stationary point charges, q_1 and q_2 , separated by a distance R , they exert a mechanical force on each other.

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{R^2} \hat{\mathbf{a}}_R \quad (1)$$

If we want to know the **Electric Field Intensity** ($\mathbf{E}$) created by q_1 alone, we define it as the force that a hypothetical +1 Coulomb "test charge" would experience if placed at a distance R . Dividing the force by $q_2 = 1$, we get the foundational equation for the electric field of a point charge:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{R^2} \hat{\mathbf{a}}_R \quad (2)$$

1.2 The Principle of Linear Superposition

What happens if you have three, four, or ten point charges? The universe follows a beautiful rule called **Linear Superposition**. The total electric field at any point in space is simply the vector sum of the individual electric fields created by each charge as if the other charges didn't even exist.

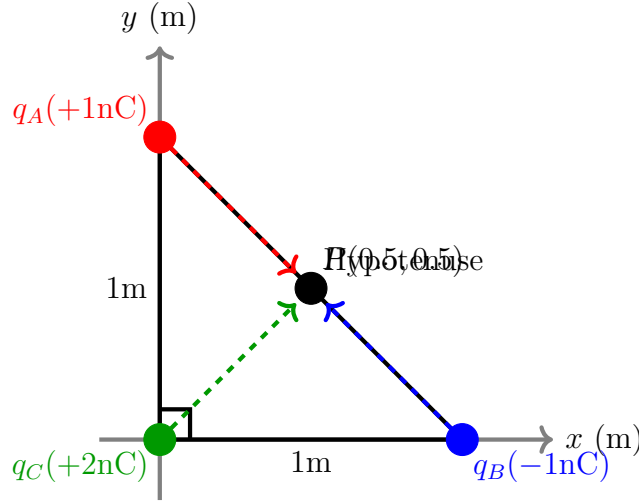
$$\mathbf{E}_{\text{total}} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \cdots + \mathbf{E}_n \quad (3)$$

1.3 Application: The Right Isosceles Triangle (PYQ 2023)

Let us apply this immediately to a classic university exam problem.

The Problem: A right isosceles triangle of side 1m has charges +1nC, +2nC and -1nC arranged on its vertices. Find the electric field at the mid-point of the hypotenuse connecting the +1nC and -1nC charges.

TikZ Diagram: Discrete Superposition



Step 1: Distance Vectors We place the right angle at the origin $C(0,0)$. The other two vertices are at $A(0,1)$ and $B(1,0)$. The midpoint of the hypotenuse is exactly at $P(0.5, 0.5)$.

We calculate the distance vectors from each charge to point P :

$$\mathbf{R}_A = (0.5 - 0)\hat{\mathbf{i}} + (0.5 - 1)\hat{\mathbf{j}} = 0.5\hat{\mathbf{i}} - 0.5\hat{\mathbf{j}} \quad (4)$$

$$\mathbf{R}_B = (0.5 - 1)\hat{\mathbf{i}} + (0.5 - 0)\hat{\mathbf{j}} = -0.5\hat{\mathbf{i}} + 0.5\hat{\mathbf{j}} \quad (5)$$

$$\mathbf{R}_C = (0.5 - 0)\hat{\mathbf{i}} + (0.5 - 0)\hat{\mathbf{j}} = 0.5\hat{\mathbf{i}} + 0.5\hat{\mathbf{j}} \quad (6)$$

Because P is the midpoint of an isosceles right triangle, it is geometrically equidistant from all three vertices! The magnitude of all three distance vectors is identical:

$$|\mathbf{R}| = \sqrt{0.5^2 + 0.5^2} = \sqrt{0.5} = \frac{1}{\sqrt{2}} \text{ meters} \quad (7)$$

Step 2: Calculate Individual Fields Using Coulomb's law in vector form: $\mathbf{E} = k \frac{q}{|\mathbf{R}|^3} \mathbf{R}$. The term $\frac{1}{|\mathbf{R}|^3}$ evaluates to $\frac{1}{(\sqrt{0.5})^3} = 2\sqrt{2}$.

For charge A (+1 nC):

$$\mathbf{E}_A = (9 \times 10^9)(1 \times 10^{-9})(2\sqrt{2})(0.5\hat{\mathbf{i}} - 0.5\hat{\mathbf{j}}) \quad (8)$$

$$\mathbf{E}_A = 18\sqrt{2}(0.5\hat{\mathbf{i}} - 0.5\hat{\mathbf{j}}) = 9\sqrt{2}\hat{\mathbf{i}} - 9\sqrt{2}\hat{\mathbf{j}} \quad (9)$$

For charge B (-1 nC):

$$\mathbf{E}_B = (9 \times 10^9)(-1 \times 10^{-9})(2\sqrt{2})(-0.5\hat{\mathbf{i}} + 0.5\hat{\mathbf{j}}) \quad (10)$$

$$\mathbf{E}_B = -18\sqrt{2}(-0.5\hat{\mathbf{i}} + 0.5\hat{\mathbf{j}}) = 9\sqrt{2}\hat{\mathbf{i}} - 9\sqrt{2}\hat{\mathbf{j}} \quad (11)$$

Notice that $\mathbf{E}_A$ and $\mathbf{E}_B$ are identical. A pushes the test charge down-right, and B pulls it down-right. They work together!

For charge C (+2 nC):

$$\mathbf{E}_C = (9 \times 10^9)(2 \times 10^{-9})(2\sqrt{2})(0.5\hat{\mathbf{i}} + 0.5\hat{\mathbf{j}}) \quad (12)$$

$$\mathbf{E}_C = 36\sqrt{2}(0.5\hat{\mathbf{i}} + 0.5\hat{\mathbf{j}}) = 18\sqrt{2}\hat{\mathbf{i}} + 18\sqrt{2}\hat{\mathbf{j}} \quad (13)$$

Step 3: Total Superposition Summing the vectors together:

$$\mathbf{E}_{\text{total}} = (9\sqrt{2} + 9\sqrt{2} + 18\sqrt{2})\hat{\mathbf{i}} + (-9\sqrt{2} - 9\sqrt{2} + 18\sqrt{2})\hat{\mathbf{j}} \quad (14)$$

$$\mathbf{E}_{\text{total}} = 36\sqrt{2}\hat{\mathbf{i}} + 0\hat{\mathbf{j}} \quad (15)$$

The magnitude is $36\sqrt{2} \approx 50.91$ V/m, pointing perfectly along the positive x-axis.

2 Continuous Charge Distributions

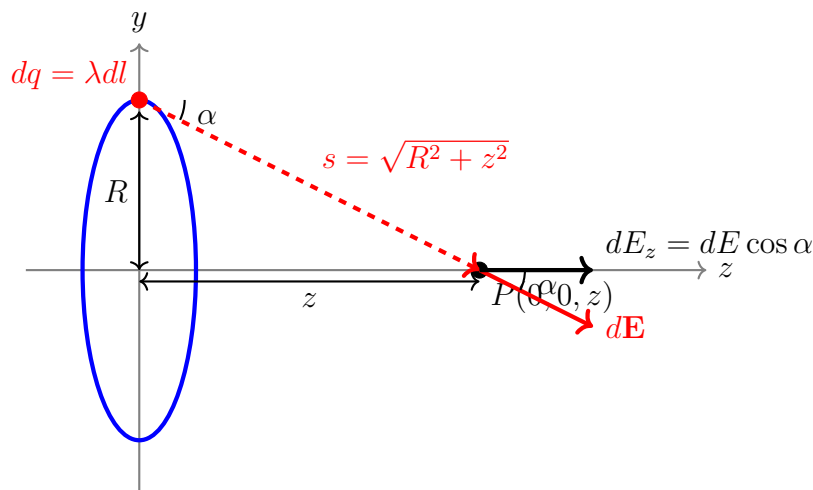
In the real world, charges are rarely isolated dots. They are smeared across wires (line charge, λ), surfaces (surface charge, σ), or volumes (volume charge, ρ_v).

To solve these, we use calculus. We slice the object into an infinite number of tiny, discrete point charges (dq), use Coulomb's Law for each one, and integrate them all together.

2.1 The Uniform Ring of Charge

The Goal: Find the electric field on the central z-axis of a very thin ring of radius R carrying a uniform linear charge density λ (C/m).

TikZ Diagram: Field of a Ring Charge



Step 1: The Differential Charge We take a tiny slice of the ring with arc length dl . The charge inside this slice is $dq = \lambda dl$.

Step 2: The Differential Field The distance s from this tiny slice at the top of the ring to the observation point P on the z-axis is found using the Pythagorean theorem: $s = \sqrt{R^2 + z^2}$. The magnitude of the differential electric field at P is:

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{s^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{R^2 + z^2} \quad (16)$$

Step 3: Symmetry and Cancellation This $d\mathbf{E}$ vector points down and to the right. It has a z-component and a vertical y-component. However, the ring is perfectly symmetric. For the tiny slice at the top of the ring, there is an identical slice at the bottom of the ring. The slice at the bottom creates a field pointing UP and to the right.

When you add them together, the vertical (transverse) components perfectly cancel out! Only the horizontal z-component survives.

Step 4: The Surviving Component The surviving z-component is $dE_z = dE \cos \alpha$. From the geometric triangle in the diagram, $\cos \alpha = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{z}{s} = \frac{z}{\sqrt{R^2 + z^2}}$.

Substitute this into our field equation:

$$dE_z = \left(\frac{1}{4\pi\epsilon_0} \frac{\lambda dl}{R^2 + z^2} \right) \left(\frac{z}{\sqrt{R^2 + z^2}} \right) \quad (17)$$

$$dE_z = \frac{\lambda z dl}{4\pi\epsilon_0 (R^2 + z^2)^{3/2}} \quad (18)$$

Step 5: Integration To find the total field, we integrate around the entire circular ring. The variables λ , z , and R are all constant as we move around the ring. The only thing that changes is dl .

$$E_{\text{ring}} = \int_{\text{ring}} dE_z = \frac{\lambda z}{4\pi\epsilon_0 (R^2 + z^2)^{3/2}} \int dl \quad (19)$$

The integral of dl around a full circle is simply the circumference, $2\pi R$.

$$E_{\text{ring}} = \frac{\lambda z (2\pi R)}{4\pi\epsilon_0 (R^2 + z^2)^{3/2}} \quad (20)$$

$$E_{\text{ring}} = \frac{\lambda R z}{2\epsilon_0 (R^2 + z^2)^{3/2}} \hat{\mathbf{z}} \quad (21)$$

2.2 The Uniform Disc of Charge

The Goal: Find the electric field on the central z-axis of a solid flat disc of radius R_D carrying a uniform surface charge density σ (C/m²).

Scaffolded Logic: We don't need to start from scratch. A solid disc is simply an infinite collection of concentric, infinitely thin rings layered tightly together from radius $r = 0$ out to radius $r = R_D$. We will use our newly derived Ring formula as our building block!

Step 1: The Differential Ring Consider a single differential ring located at radius r , with a tiny radial thickness dr . The area of this thin ring is its circumference times its thickness: $dA = 2\pi r dr$. The total charge on this specific differential ring is:

$$dq_{\text{ring}} = \sigma dA = \sigma(2\pi r dr) \quad (22)$$

Step 2: The Differential Field We take the Ring formula we just derived (Equation 21). We replace the total ring charge ($\lambda 2\pi R$) with our new differential charge dq_{ring} , and we swap the fixed ring radius R for our variable integration radius r :

$$dE_z = \frac{z dq_{\text{ring}}}{4\pi\epsilon_0(r^2 + z^2)^{3/2}} \quad (23)$$

$$dE_z = \frac{z(\sigma 2\pi r dr)}{4\pi\epsilon_0(r^2 + z^2)^{3/2}} \quad (24)$$

$$dE_z = \frac{\sigma z}{2\epsilon_0} \frac{r}{(r^2 + z^2)^{3/2}} dr \quad (25)$$

Step 3: Integration over the Disc To find the total field, we integrate this differential ring field from the dead center of the disc ($r = 0$) out to the physical edge of the disc ($r = R_D$).

$$E_{\text{disc}} = \int_0^{R_D} \frac{\sigma z}{2\epsilon_0} r(r^2 + z^2)^{-3/2} dr \quad (26)$$

We pull out the constants and set up a standard u -substitution. Let $u = r^2 + z^2$. Then the derivative is $du = 2r dr$, meaning $r dr = \frac{du}{2}$. The bounds of integration change from $[0, R_D]$ to $[z^2, R_D^2 + z^2]$.

$$E_{\text{disc}} = \frac{\sigma z}{2\epsilon_0} \int_{z^2}^{R_D^2 + z^2} u^{-3/2} \frac{du}{2} \quad (27)$$

$$= \frac{\sigma z}{4\epsilon_0} \left[\frac{u^{-1/2}}{-1/2} \right]_{z^2}^{R_D^2 + z^2} \quad (28)$$

$$= -\frac{\sigma z}{2\epsilon_0} \left[\frac{1}{\sqrt{R_D^2 + z^2}} - \frac{1}{\sqrt{z^2}} \right] \quad (29)$$

Assuming we are strictly looking at the positive z-axis ($z > 0$), we know $\sqrt{z^2} = z$. Distributing the negative sign and pulling z inside the brackets:

$$E_{\text{disc}} = \frac{\sigma}{2\epsilon_0} \left[\frac{z}{z} - \frac{z}{\sqrt{R_D^2 + z^2}} \right] \quad (30)$$

$$E_{\text{disc}} = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{R_D^2 + z^2}} \right] \hat{\mathbf{z}} \quad (31)$$

Infinite Sheet Approximation: Notice what happens if you make the disc infinitely large ($R_D \rightarrow \infty$). The fraction term $\frac{z}{\infty}$ evaluates to zero. The equation beautifully collapses to $E = \frac{\sigma}{2\epsilon_0}$, which is the exact, well-known formula for an infinite sheet of charge!

3 Gauss's Law: The Ultimate Symmetry Shortcut

Calculating the electric field of complex shapes using Coulomb's law integrals is brutal. However, if the shape possesses perfect symmetry (like a sphere, a long cylinder, or an infinite flat plane), we can bypass the integrals entirely using **Gauss's Law**.

3.1 The Integral Form

Gauss's Law states that the total electric flux (Ψ) bursting outward through any closed, imaginary 3D surface (a Gaussian Surface) is exactly and identically equal to the total free charge enclosed within that volume.

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}} \quad (32)$$

3.2 The Differential Point Form (Derivation)

While the integral form handles macroscopic shapes, we often need to know what is happening at a specific, microscopic point in space.

Step 1: We mathematically express the total enclosed charge Q_{enc} as the volume integral of the continuous volume charge density ρ_v over the enclosed volume V :

$$Q_{\text{enc}} = \iiint_V \rho_v dv \quad (33)$$

Substitute this directly back into Gauss's Law:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \iiint_V \rho_v dv \quad (34)$$

Step 2: We invoke the **Divergence Theorem** exactly on the left-hand side of Equation 34. The Divergence Theorem allows us to convert the closed surface integral of the flux density vector field $\mathbf{D}$ into a volume integral of the spatial divergence of that exact same field:

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{D}) dv \quad (35)$$

Step 3: Substitute this divergence conversion back into the core equation:

$$\iiint_V (\nabla \cdot \mathbf{D}) dv = \iiint_V \rho_v dv \quad (36)$$

We mathematically merge the two integrals:

$$\iiint_V [\nabla \cdot \mathbf{D} - \rho_v] dv = 0 \quad (37)$$

Because the initial physical choice of the bounded volume V was entirely arbitrary, this integrated relationship must hold rigorously true for any conceivable volume, regardless of its size or shape. The only mathematical way a volume integral can evaluate to zero for absolutely all possible volume boundaries is if the internal integrand itself evaluates identically to zero everywhere in space.

$$\begin{aligned}\nabla \cdot \mathbf{D} - \rho_v &= 0 \\ \nabla \cdot \mathbf{D} &= \rho_v\end{aligned}\tag{38}$$

This is the very first Maxwell Equation. It dictates that the outward "spraying" (divergence) of electric flux density at any microscopic point is directly and solely caused by the presence of a volume charge density at that exact coordinate.

4 Electric Potential and Orthogonality

4.1 Defining Electric Potential

Electric field intensity ($\mathbf{E}$) deals with force vectors. Electric Potential (V) deals with scalar energy.

The potential difference between two points, A and B, is defined as the external mechanical work required to physically push a positive unit test charge from point A to point B against the opposing electric field.

$$V_{AB} = - \int_A^B \mathbf{E} \cdot d\mathbf{l} \quad (39)$$

The negative sign physically represents that you must do work *against* the natural direction of the electric field to increase the potential energy of the charge (just like you must push against gravity to lift a boulder up a hill).

4.2 PYQ Derivation: Proving Orthogonality

The Problem: Prove that electric flux lines and equipotential surfaces are always normal (perpendicular) to each other.

Step 1: Define the Equipotential Surface An equipotential surface is a geometric 3D manifold along which the scalar potential $V(x, y, z)$ is perfectly constant. If you move a test charge along any path strictly lying upon this surface, absolutely no work is done, because the potential is not changing. Let $d\mathbf{l}$ be an infinitesimally small displacement vector tangent to this surface.

$$dV = 0 \quad \text{for any displacement } d\mathbf{l} \text{ on the surface.} \quad (40)$$

Step 2: The Gradient Relationship The fundamental mathematical relationship linking the vector electric field and the scalar potential field is defined by the negative spatial gradient:

$$\mathbf{E} = -\nabla V \quad (41)$$

From multivariate calculus, the total differential change dV of a scalar function $V(x, y, z)$ along an arbitrary spatial displacement vector $d\mathbf{l}$ is defined by the exact differ-

ential formula:

$$\begin{aligned}
dV &= \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz \\
dV &= \left(\frac{\partial V}{\partial x}\hat{\mathbf{i}} + \frac{\partial V}{\partial y}\hat{\mathbf{j}} + \frac{\partial V}{\partial z}\hat{\mathbf{k}} \right) \cdot (dx\hat{\mathbf{i}} + dy\hat{\mathbf{j}} + dz\hat{\mathbf{k}}) \\
dV &= (\nabla V) \cdot d\mathbf{l}
\end{aligned} \tag{42}$$

Step 3: Synthesis We systematically substitute the gradient relationship defined in Equation 41 directly into our total differential expression:

$$\begin{aligned}
dV &= (-\mathbf{E}) \cdot d\mathbf{l} \\
dV &= -\mathbf{E} \cdot d\mathbf{l}
\end{aligned} \tag{43}$$

Now, we impose the strict physical boundary condition we defined in Equation 40: we explicitly confine our displacement vector $d\mathbf{l}$ to lie entirely upon the constant equipotential surface. Therefore, the differential potential change dV must evaluate identically to zero.

$$0 = -\mathbf{E} \cdot d\mathbf{l} \implies \mathbf{E} \cdot d\mathbf{l} = 0 \tag{44}$$

The dot product of two geometric vectors $\mathbf{A}$ and $\mathbf{B}$ is mathematically defined as $|\mathbf{A}||\mathbf{B}|\cos(\theta)$, where θ represents the interior angle between them.

$$|\mathbf{E}||d\mathbf{l}|\cos(\theta) = 0 \tag{45}$$

Assuming a real electric field exists ($|\mathbf{E}| \neq 0$) and we execute a non-zero displacement ($|d\mathbf{l}| \neq 0$), the only mathematical way for this product to evaluate to zero is if the trigonometric cosine of the angle between them is exactly zero:

$$\begin{aligned}
\cos(\theta) &= 0 \\
\implies \theta &= 90^\circ \quad \left(\text{or } \frac{\pi}{2} \text{ radians} \right)
\end{aligned} \tag{46}$$

This rigorously proves that electric flux lines (defined by $\mathbf{E}$) must **always** intersect equipotential surfaces at perfect, orthogonal right angles.

5 Phase 1 Summary and Exam Checklist

Congratulations! You have successfully mastered the physics of stationary charges. You have moved from calculating discrete point charges to integrating continuous smeared charges, and finally to bypassing integrals entirely using Gauss's symmetry.

Before advancing to Phase 2 (Boundaries & Energy), ensure you can confidently check off every item on this list:

- ☐ I can fully reproduce the vector math to calculate the electric field of a right isosceles triangle of charges without making sign errors on the distance vectors.
- ☐ I can derive the exact electric field formula for a continuous uniform ring of charge from first principles. I understand why the vertical components perfectly cancel.
- ☐ I can derive the electric field formula for a solid disc of charge by integrating a series of differential rings. I know the u-substitution required.
- ☐ I can formally derive the differential point-form of Gauss's Law ($\nabla \cdot \mathbf{D} = \rho_v$) by merging the integral form of Gauss's Law with the Divergence Theorem.
- ☐ I can mathematically prove that electric flux lines and equipotential surfaces cross at exactly 90 degrees using the gradient dot product relationship.

Once this checklist is complete, you are fully prepared to tackle the physics of what happens when these fields crash into solid matter in **Phase 2: Boundaries, Energy, & Conductors**.